

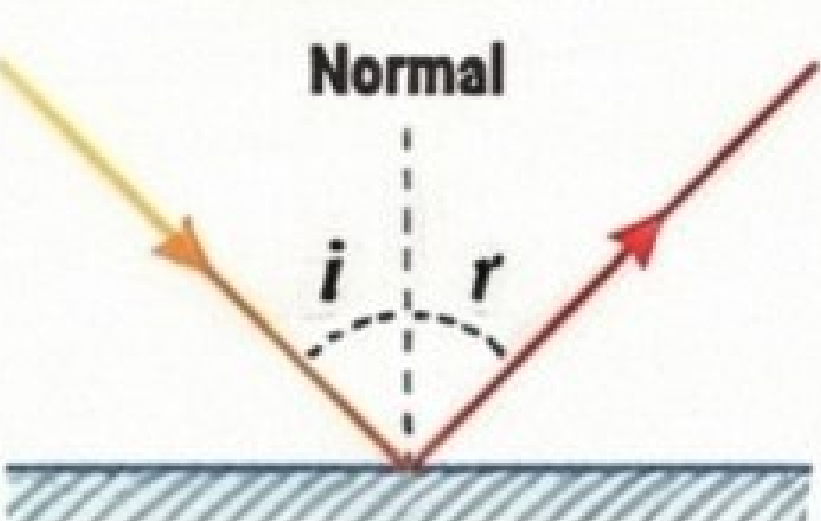
Ray Optics: A Visual Guide to Light

1. Fundamentals of Light

What is Optics?

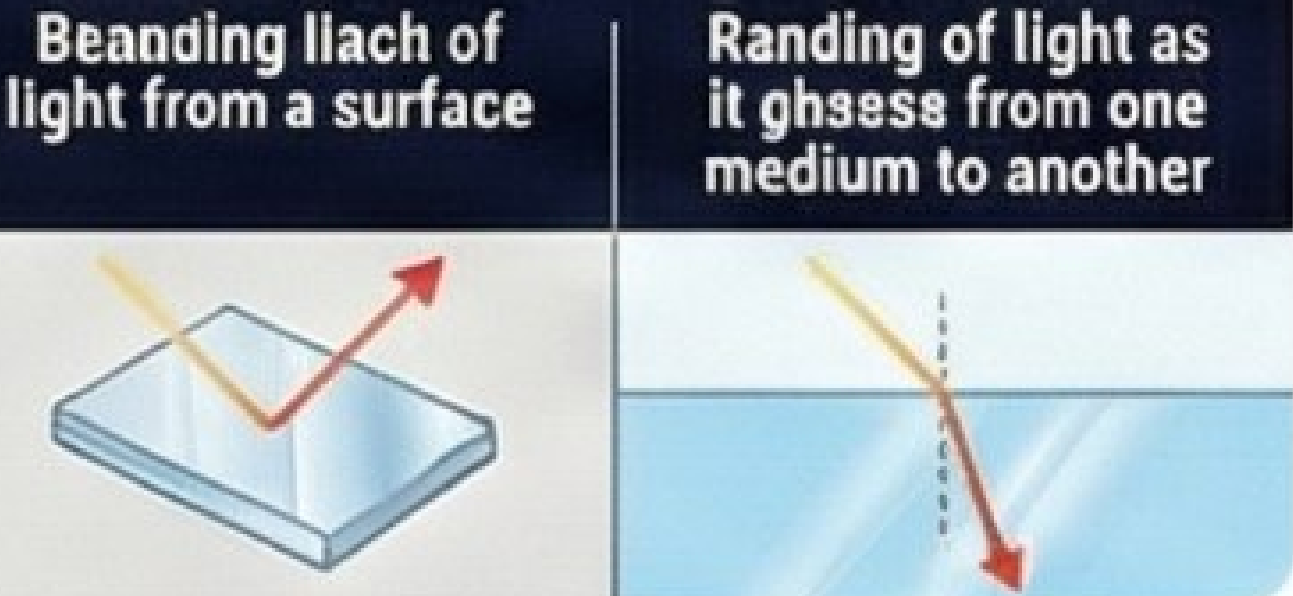
The branch of physics that studies the behavior and properties of light, and the construction of instruments that use or detect it.

Laws of Reflection



- 1. The angle of incidence (i) equals the angle of reflection (r).
- 2. The incident ray, reflected ray, and the normal all lie in the same plane.

Reflection vs. Refraction



- 1. The incident ray, refracted ray, and the normal all lie in the same plane.
- 2. Snell's Law: The ratio $\sin(i)/\sin(r)$ is a constant, known as the refractive index (n).

2. Mirrors & Images

Real vs. Virtual Images



Real images can be formed on a screen and are always inverted.



Virtual images cannot be formed on a screen and are always erect.

The Mirror Formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Magnification (m):

$$m = \frac{\text{Height of Image}}{\text{Height of Object}} = -\frac{v}{u}$$

A negative 'm' indicates a real, inverted image.

Mirror Types	Concave Mirror	Convex Mirror
Action on Light	Converges (focuses) light	Diverges (spreads) light
Image Type	Mostly Real & inverted (one case of Virtual & Erect)	Always Virtual, Erect & Diminished
Common Uses	Shaving mirrors, dentist's tools, headlights	Vehicle side-view mirrors, security mirrors
Focal Length	Negative	Positive

3. Refraction, Lenses & Total Internal Reflection (TIR)

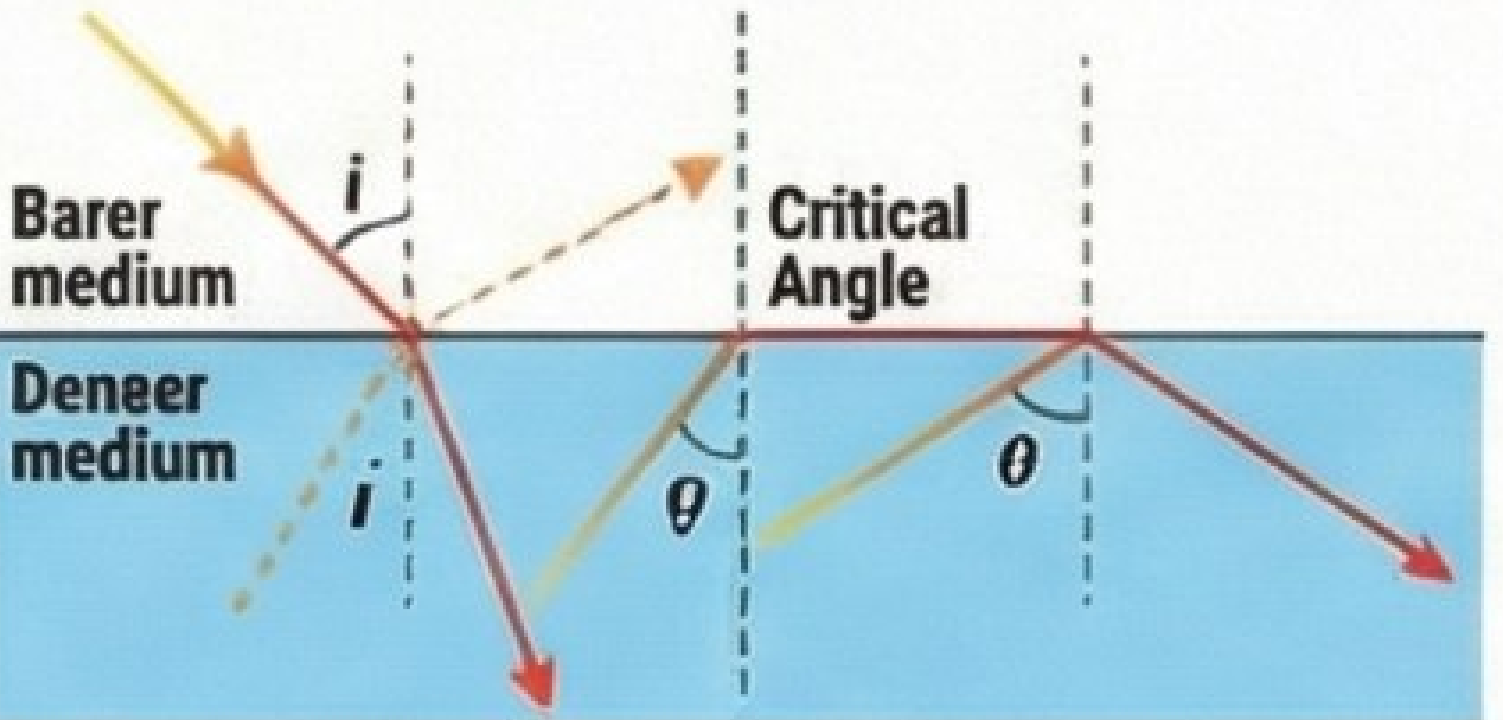
Refractive Index (n)

A measure of how much light slows down in a medium.

$$n = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

Total Internal Reflection (TIR)

- 1. Light travels from a denser to a rarer medium.
- 2. Angle of incidence is greater than the critical angle.



Total Internal Reflection (TIR)

The Lens Formula:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Lens Maker's Formula:

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

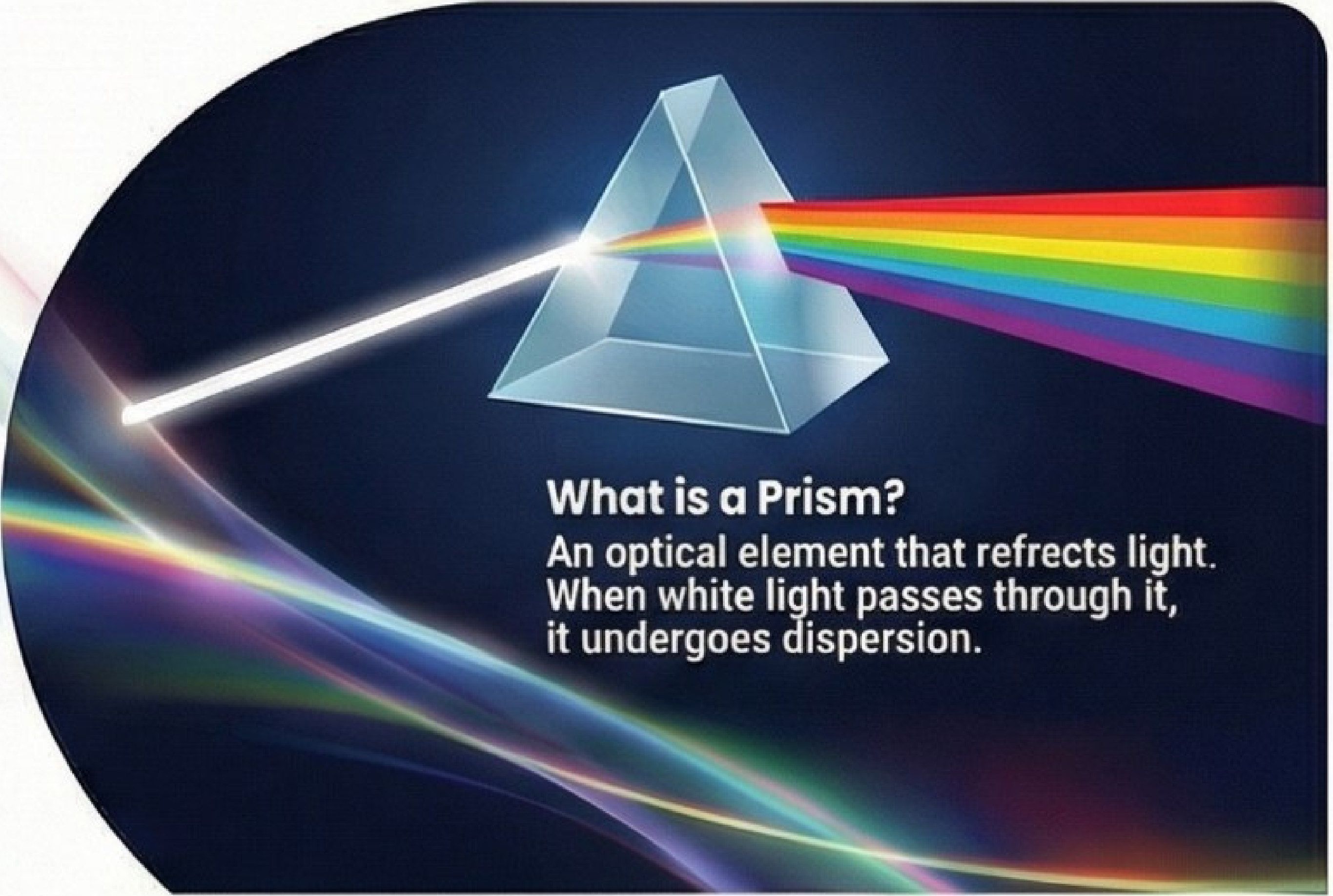
Power of a Lens (P):

The ability of a lens to converge or diverge light, measured in Diopters (D).

$P = 1/f$ (where f is in meters).
Positive for convex, negative for concave.

Lens Types	Convex Lens	Concave Lens
Action on Light	Converges (focuses) light	Diverges (spreads) light
Shape	Thicker in the middle, thinner at the edges	Thinner in the middle, thicker at the edges
Image Type	Mostly Real & Inverted (one case of Virtual & Erect)	Always Virtual, Erect & Diminished
Common Uses	Magnifying glasses, microscopes, cameras	Eyeglasses for nearsightedness, telescopes

4. The Prism

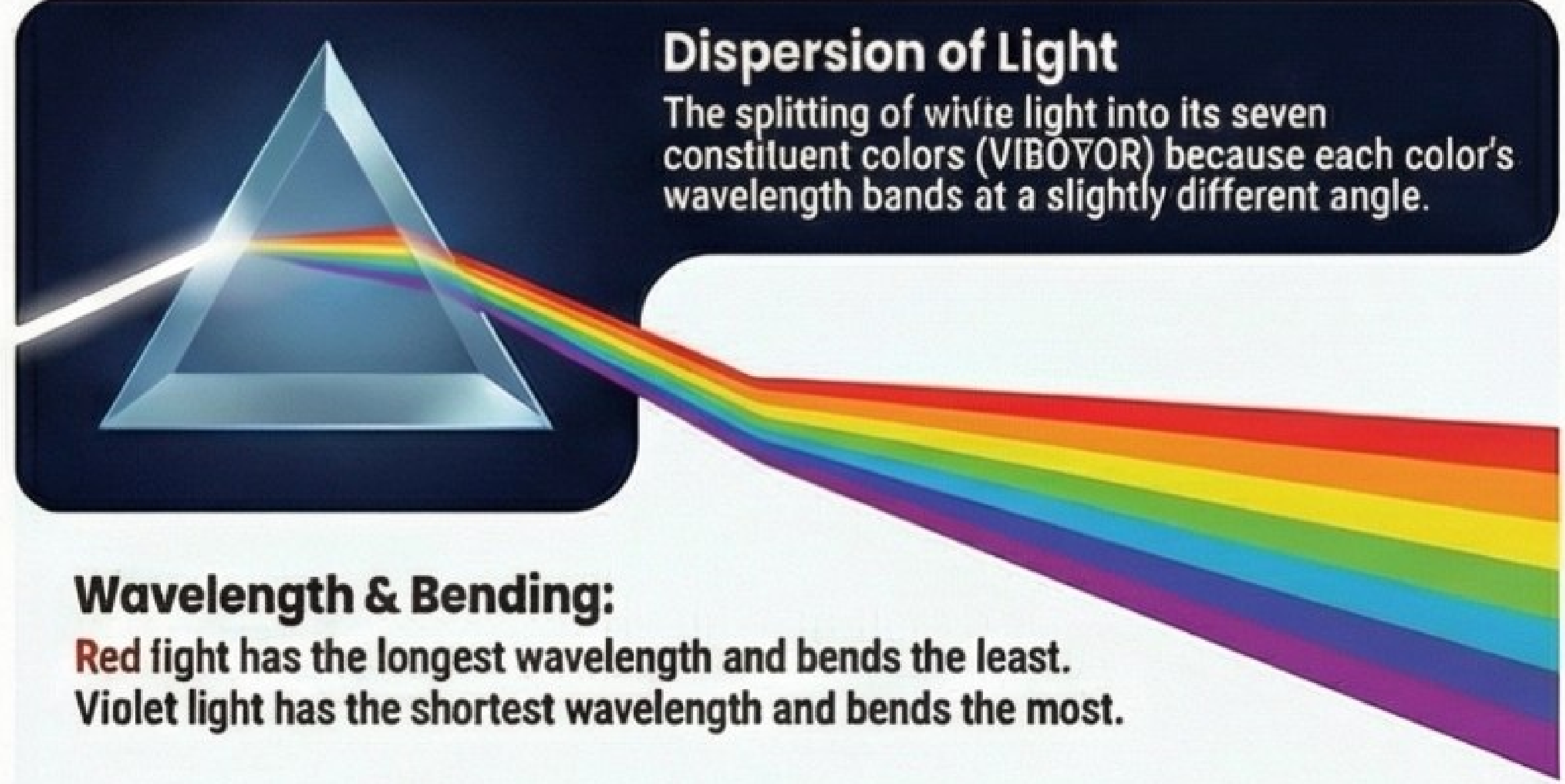


What is a Prism?

An optical element that refracts light. When white light passes through it, it undergoes dispersion.

Dispersion of Light

The splitting of white light into its seven constituent colors (VIBGYOR) because each color's wavelength bends at a slightly different angle.



Wavelength & Bending:

Red light has the longest wavelength and bends the least.
Violet light has the shortest wavelength and bends the most.

The Prism Formula

Calculates the refractive index (n) of the prism's material using the Angle of Prism (A) and the Angle of Minimum Deviation (δ_m):

$$n = \frac{\sin((A + \delta_m)/2)}{\sin(A/2)}$$